

UTM TENSILE REPORT — UTM_Test_20260817_171241

DIC gauge strain (ENGINEERING: $\Delta L/L_0$) · engineering stress (F/A₀) · anchor self-calibration

47.8 MPa

UTS

42.7 MPa

Yield σ_y

3.23 GPa

Modulus E

5.8 %

Failure ϵ_f

2457 kJ/m³

Toughness

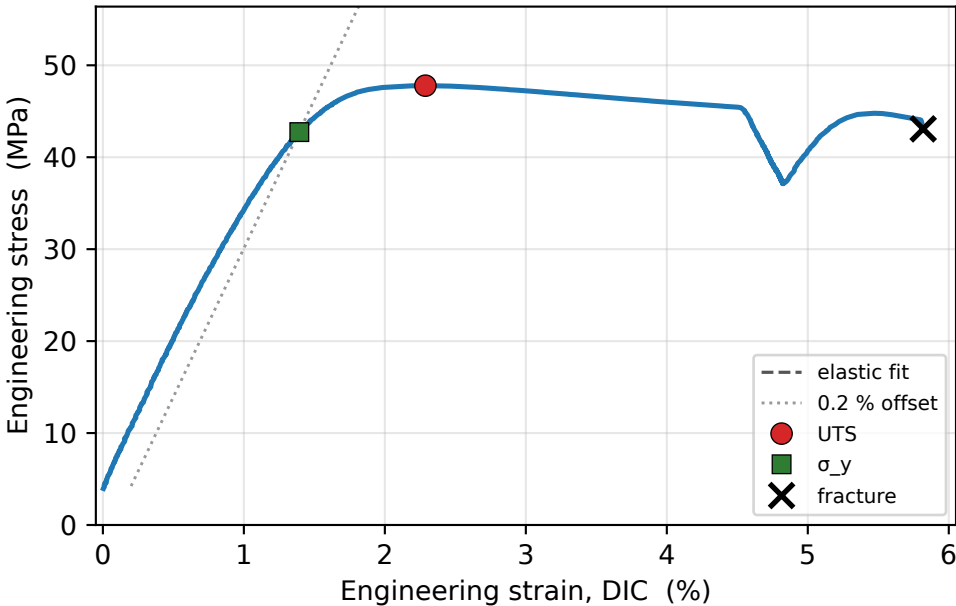
317 N

Preload anchor

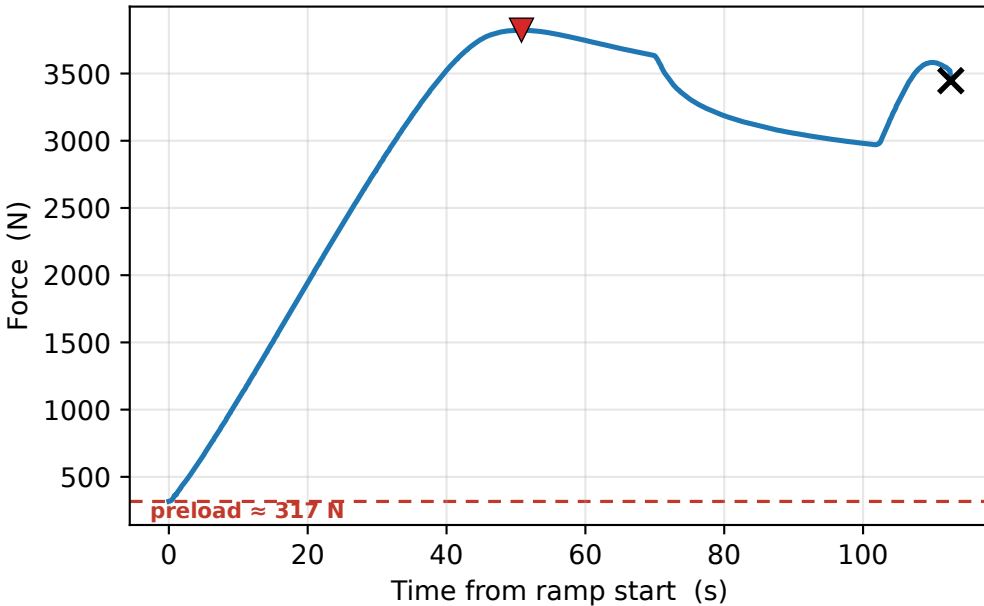
TEST SETTINGS

Specimen	UTM_Test_20260817_171241
DIC mode	Black
Preload target	300 N
Test speed	0.100 mm/s
Cross-section	80 mm ²
Gauge length L ₀	80 mm
Infill (label)	100 %
Force calib	sc -0.0065 / off 341.893
Comment	—
Test date	2026-08-17 17:09:50
Duration	178.3 s

Stress-strain



Load vs time

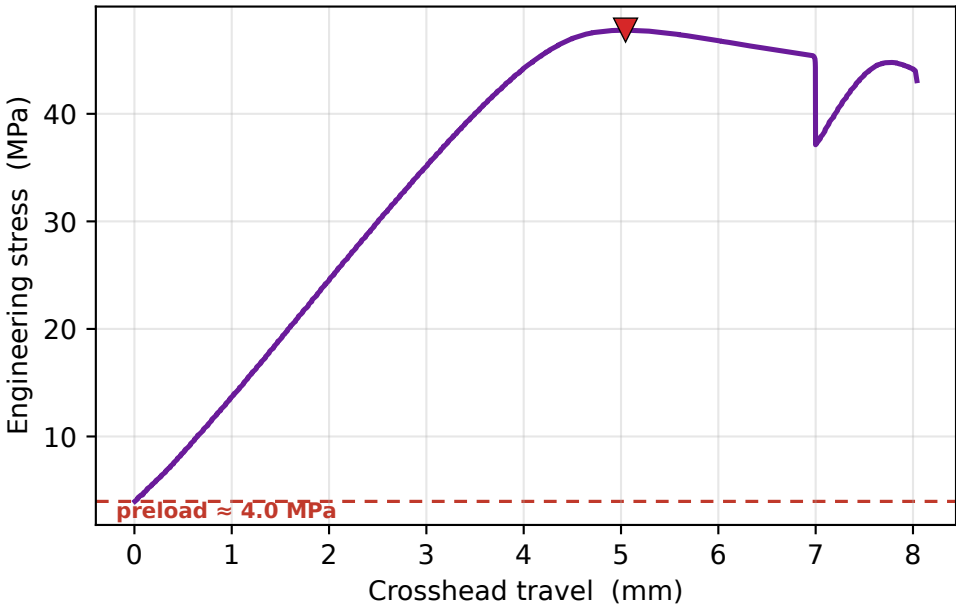


VALIDATION vs REFERENCES

prop	meas	Chacón	E-PLA k
UTS	47.8	32-60 PASS	×1.21
σ_y	42.7	30-50 PASS	×1.36
E	3.2	3-5.5 PASS	×0.89
ϵ_f	5.8 %	(range 3-7 %)	×1.37

✓ strength inside journal range (k≈1)

Stress vs crosshead



DIC strain vs crosshead — the compliance gap

